Healthcare Operations: Predictive Model for Encounter Total Claim Cost

Track C – Machine Learning Engineering | AnalystLab Africa

1. Problem Selection & Justification

Selected problem: Predict the **total claim cost** of a hospital encounter (regression).

Why feasible and operationally relevant: Target TOTAL_CLAIM_COST is present and non-null for all 27,891 encounters. Sufficient volume for supervised learning. At the point an encounter begins (or is scheduled), an estimate of expected cost supports capacity planning, financial forecasting, and early identification of potentially high-cost cases for care management.

Alternatives examined and rejected: Duration > 24 h (only ~75 positives, 0.27% + data-quality outliers); Zero payer coverage (highly determined by known PAYER field, near-trivial); 30-day repeat encounter (constructible but requires careful windows and risks treating every subsequent visit as clinical readmission).

Prediction point: Features known or knowable at the **start** of the encounter (or at scheduling time). No information generated during or after the encounter is used as a feature.

2. Target & Features

Target: TOTAL_CLAIM_COST (USD). Continuous, right-skewed (mean ~ $3.6k, median ~ $279, max > $600k).

Features available at prediction time

Feature	Type	Notes
ENCOUNTERCLASS	categorical	ambulatory, outpatient, urgentcare, emergency, wellness, inpatient
PAYER_NAME	categorical	10 payers including NO_INSURANCE
GENDER / RACE / ETHNICITY / MARITAL	categorical	Patient demographics
AGE_AT_START	numeric	Derived from BIRTHDATE + START
YEAR, MONTH, DOW, IS_WEEKEND	numeric / binary	Calendar features from START

Features deliberately excluded (leakage or unavailable): STOP, DURATION_HOURS; all procedure-level data; BASE_ENCOUNTER_COST, PAYER_COVERAGE, REASONCODE; patient identifiers, names, addresses.

Missing-data handling: MARITAL (12 missing) filled with "U". All other modelling features complete after join.

Limitations of target: Synthetic data (Synthea-style); single organisation; extreme right tail driven by rare high-intensity procedures that the model cannot see (by design). Consequently R^2 remains modest (~0.15). Model is **not** a diagnostic, clinical, or treatment system.

3. Baseline, Metrics, Validation Strategy

Baselines: DummyRegressor (mean) – primary; median predictor also reported.

Metrics: MAE (primary, dollar-interpretable), RMSE, R^2, Median Absolute Error (MedAE). Hospital finance cares about absolute dollar accuracy more than relative percentage error on low-cost visits.

Validation: Chronological split – Train START year <= 2019 (n=21,622); Test 2020–2022 (n=6,269). Operational deployment would train on historical data and score future encounters. Random/k-fold would leak future information.

4. Models Trained & Comparison

Model	Test MAE	Test RMSE	Test R^2	Test MedAE
Mean baseline	5,125	12,305	-0.002	3,598
Median baseline	3,083	12,616	-0.053	238
Ridge (linear)	4,391	11,991	0.048	2,811
Random Forest	**3,066**	11,326	0.151	1,123
Gradient Boosting	3,124	**11,279**	**0.158**	1,316

Tree ensembles clearly outperform linear models and the mean baseline on MAE and R^2. Random Forest selected as primary candidate (best MAE, competitive RMSE/R^2, interpretable feature importances). Top drivers: age, gender, encounter class (urgentcare / ambulatory / inpatient), NO_INSURANCE payer.

5. Error Analysis & Limitations

Largest absolute errors occur on rare ultra-high-cost encounters (inpatient / emergency, often > $100k). Top residual ~ $637k (true $642k, predicted ~$5k). These cases are driven by procedures invisible to the model by design. By class, inpatient has highest MAE (~$10k); ambulatory has lowest. Residuals roughly centred (mean residual ~ +11) but heavy-tailed.

Data leakage check: All features constructed exclusively from information available at START; procedure table never joined; STOP-derived fields excluded.

Further monitoring before operational use: Prediction vs actual distributions over time (drift); performance by class and payer; calibration of high-cost tail; retraining cadence; human review of predicted high-cost cases rather than automated decisions.

6. Reproducible Workflow

Source: data_prep.py (feature engineering + chronological split), train.py (baseline + 3 models), inference.py (load, validate, predict). Tests in test_pipeline.py (all passing). Models saved as joblib pipelines. Metrics and test predictions written to reports/.

```
cd ml_project/src
python train.py
python ../tests/test_pipeline.py
python inference.py --input sample.csv --model random_forest
```

7. Responsible Use Statement

This work concerns **healthcare operations** only. The model is **not** a diagnostic or treatment system. No clinical validity is claimed. Patient names, addresses and other direct identifiers are never used as features and never appear in reports or saved artefacts. The underlying data are synthetic; results must not be generalised to real clinical populations without further validation under appropriate governance.